

Decent Coaching Classes

IX CBSE

Mathematics: The World of Numbers & Exploring Algebraic Identities

Time: 1 Hour

Marks: 30

Section A - Choose the correct alternative. (6 × 1 = 6)

1) Which of the following is an irrational number?

- a) $\sqrt{16}$ b) $\sqrt{2}$ c) $\frac{4}{9}$ d) 0.333...

2) Every rational number is a:

- a) natural number b) whole number c) real number d) integer

3) The value of $(2 + \sqrt{3})(2 - \sqrt{3})$ is:

- a) 1 b) $4 - \sqrt{3}$ c) 7 d) $4 + 3$

4) Using the identity for $(x + y)^2$, we get:

- a) $x^2 + y^2$ b) $x^2 + 2xy + y^2$ c) $x^2 - 2xy + y^2$ d) $x^2 + xy + y^2$

5) $(a + b)^3$ expands to:

- a) $a^3 + b^3$ b) $a^3 + 3a^2b + 3ab^2 + b^3$ c) $a^3 - 3a^2b + 3ab^2 - b^3$ d) $a^3 + ab + b^3$

6) $x^3 - y^3 = ?$

- a) $(x - y)(x^2 + xy + y^2)$ b) $(x - y)(x^2 - xy + y^2)$ c) $(x + y)(x^2 - xy + y^2)$ d) $(x - y)(x + y)^2$

Section B - Answer the following. (4 × 2 = 8)

1) Describe briefly how $\sqrt{5}$ can be represented on the number line.

2) Simplify: $(\sqrt{3} + \sqrt{2})^2$

3) If $x + y = 5$ and $xy = 6$, find the value of $x^3 + y^3$ using a suitable identity.

4) Rationalize the denominator: $1 / (\sqrt{7} - \sqrt{5})$

Section C - Answer the following. (4 × 3 = 12)

1) Rationalize the denominator and simplify: $(3 + \sqrt{2}) / (3 - \sqrt{2})$

2) Using a suitable identity, expand: $(2x + 3y - 4z)^2$

3) Factorise using a standard identity: $27x^3 + 64y^3$

4) Evaluate 105×95 using an algebraic identity.

Section D - Answer the following. (1 × 4 = 4)

1) If $x + 1/x = 5$, find the value of $x^3 + 1/x^3$, showing all steps using a suitable identity.

— All the Best —